

Step-by-Step Installation & Run Guide

Follow these instructions to configure and execute the AgriGrow Crop Recommendation Web Portal locally on your machine.

1 Start the MySQL Database Server

This web application utilizes a MySQL database for storing prediction history and cultivation statuses.

1. Open the **XAMPP Control Panel** on your computer.
2. Locate the **MySQL** row and click the **Start** button.
3. Ensure the module shows a green background with port **3306** active.

MYSQL DATABASE DETAILS

Database Engine: **MySQL (via XAMPP)**

Database Schema Name: **crop_recommendation**

Port: **3306** (localhost)

2 Open Terminal in Project Directory

Open command line terminal (PowerShell, Command Prompt, or terminal inside VS Code) and navigate to the project directory:

```
cd "E:\INTERN PROJECT\project"
```

3 Activate the Python Virtual Environment

A virtual environment containing the necessary dependencies (Django, Pandas, Scikit-learn, joblib, PyMySQL) is already configured in the folder.

Run the activation script corresponding to Windows PowerShell/Command Prompt:

```
.venv\Scripts\activate
```

Once activated, you should see `(.venv)` prepended to your command line prompt.

4 Execute Database Migrations

Sync Django's database schema models with your active MySQL database:

```
python manage.py migrate
```

Note: If database is already initialized and tables exist, it will output "No migrations to apply".

5 Launch Django Development Server

Start the local server by executing:

```
python manage.py runserver
```

The console will log startup text and indicate that the application is live at

```
http://127.0.0.1:8000/ .
```

6 Open the Web Application

1. Open your web browser (Edge, Chrome, Firefox).
2. Enter the following address in the URL bar: <http://127.0.0.1:8000/>
3. You will be redirected to the sign-in portal.

7 Administrator Login

If the database tables are empty, a default superuser account will automatically be created on your first login load. Access the administrator workspace using these credentials:

DEFAULT ADMIN CREDENTIALS

Username: **admin**

Password: **adminpassword**

8 Database Analytics & Visualizations

*
Select

**Crop
Database**

from the sidebar to inspect dataset chemical balances and climate metrics. *
Select

**ML
Analytics**

to view decision tree model weights, user prediction histories, and validation confusion matrices.